Investigating IR Radiation

Exam code: 0625 & 0972

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Investigating thermal radiation

Aims of the experiment

- The aim of the experiment is to investigate how the amount of infrared radiation absorbed or radiated by a surface depends on the nature of that surface

Variables

- **Independent variable** = Colour
- **Dependent variable** = Temperature
- Control variables:
  - Identical flasks (except for their colour)
  - Same amounts of hot water
  - Same starting temperature of the water
  - Same time interval

Equipment

Equipment List

Equipment	Purpose
Heatproof mat	To protect surfaces and reduce heat loss
Stop watch	To measure time taken for cooling
Kettle	To boil water
4 thermometers	To measure the water temperature in each flask
Flasks painted different colours (black, dull grey, white, silver)	To investigate the heat loss of different colours

- **Resolution** of measuring equipment:
  - Thermometer = 1°C
  - Stopwatch = 0.01 s

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Different coloured beakers for investigating infrared radiation apparatus

1. Set up the four identical flasks painted in different colours: black, grey, white and silver
2. Fill the flasks with hot water, ensuring the measurements start from the same initial temperature
3. Note the starting temperature, then measure the temperatures at regular intervals, e.g. every 30 seconds for 10 minutes

Results

Example results table

TIME / mins	BLACK / °C	WHITE / °C	DULL GREY / °C	SILVER / °C
0.5				
1.0				
1.5				
2.0				
2.5				
3.0				
3.5				
4.0				
4.5				
5.0				
5.5				
6.0				
6.5				
7.0				

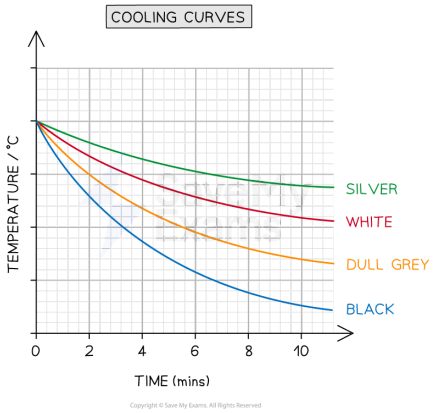
Analysis of results

- All objects emit **infrared radiation**, but the hotter an object is, the more infrared waves are emitted
- The intensity (and wavelength) of the emitted radiation depends on:
  - The **temperature** of the body (hotter objects emit more thermal radiation)
  - The **surface area** of the body (a larger surface area allows more radiation to be emitted)
  - The **colour** of the surface

- Most of the energy lost from the beakers will be **by heating** due to **conduction** and **convection**
  - This will be **equal** for each beaker, as colour does not affect energy transferred by conduction and convection

flask, plot a graph of **temperature** on the y-axis against **time** on the x-axis and draw curves of best fit

- The expected results are shown on the graph below:



Example graph of the expected results for the different coloured beakers

Evaluating the experiments

Systematic errors

- Make sure the starting temperature of the water is the same for each material since this will cool very quickly
- It is best to do this experiment in pairs to coordinate starting the stopwatch and immersing the thermometer
- Use a data logger connected to a digital thermometer to get more accurate readings

Random errors

- Make sure the hole for the thermometer isn't too big, otherwise thermal energy will escape through the hole
- Take repeated readings for each coloured flask
- Read the values on the thermometer at eye level, to avoid parallax error

Safety considerations

- Safety goggles should be worn when filling the flasks with boiling water
- Keep water away from all electrical equipment
- Make sure not to touch the hot water directly
  - Run any burns immediately under cold running water for at least 5 minutes
- Do not overfill the kettle
- Make sure all the equipment is in the middle of the desk, and not at the end to avoid knocking over the beakers
- Carry out the experiments only whilst standing, in order to react quickly to any spills or burns

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